Karen Deng

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SUMMARY

Computing and Software Systems student at The University of Melbourne with experience building full-stack applications and integrating LLMs for automation and data analysis. Built production systems across web platforms and command-line tools, and contributed to consulting projects delivering strategy frameworks and technical recommendations for real-world clients.

EDUCATION

Bachelor of Science in Computing and Software Systems Mar, 2024 - Oct, 2027
The University of Melbourne

- WAM: 80 (First Class Honours)
- Relevant Coursework: Data Structures and Algorithms, Database Systems, Foundations of Algorithms, Foundations of Computing

RELEVANT PROJECT WORK

AI Medical Case Simulator (devpost.com/software/medcase) | *SB Hacks XI* Jan, 2026
React, TypeScript, Node.js, Express, FastAPI, OpenAI

- Created an interactive medical training platform enabling students to practice patient diagnosis through AI-powered clinical interviews and treatment workflows
- Integrated OpenAI's LLM and Deepgram text-to-speech via FastAPI microservice to simulate realistic patient responses with voice output, creating an immersive clinical training experience
- Engineered in-memory session management to handle concurrent user simulations, tracking conversation history, diagnostic attempts, and scenario progression state

NLP-Driven Journal Knowledge Graph (moire-xi.vercel.app) Dec, 2025 - Jan, 2026
Next.js, React, Prisma ORM, PostgreSQL, Clerk, OpenAI

- Built a journaling platform with automated topic extraction and sentiment analysis to help users discover patterns across entries, generating LLM-powered summaries with contextual insights for each theme
- Designed a semantic knowledge graph using vector embeddings to identify and canonicalise related topics, creating dynamic connections between journal themes
- Developed an interactive D3 force-directed graph visualisation displaying topic relationships with weighted edges based on co-occurrence frequency

Python Developer Jul, 2025 - Aug, 2025
DSCubed Winter Program - University of Melbourne, Melbourne, Australia

- Built an AI-powered, modular command-line assistant, integrating OpenAI with secure .env configuration, async function calling, and a dynamic tool system using modern Python best practices
- Reduced new member onboarding time from 1+ hours to 15 minutes by automating account creation, permission setup, and resource allocation using n8n workflows

OTHER EXPERIENCE

Project Consultant Feb, 2025 - Jul, 2025
180 Degrees Consulting - University of Melbourne, Melbourne, Australia

- Delivered a 12-week project for Creative Australia, co-authoring a 30+ page impact evaluation framework with KPIs, methodologies, and data collection strategies to assess climate initiatives in the arts sector

Events Officer Feb, 2025 - Jul, 2025
AIESEC in Melbourne, Melbourne, Australia

- Managed end-to-end logistics and budgeting for events with up to 100 attendees

TECHNICAL SKILLS

- Languages: Python, JavaScript, SQL, C
- Frameworks/Libraries: React, Next.js, Node.js, Express, FastAPI, Prisma ORM, D3.js
- Technologies: PostgreSQL, OpenAI API, Git, Clerk (Auth), RESTful APIs